

Hybrid Cloud Enterprise WAN Documentation

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1 Introduction

This report details the design and setup of an enterprise network built in Cisco Packet Tracer. The network connects a local branch office to a secure Private Cloud environment through a core regional network.

To secure communication over the public ISP backbone, we deployed a Site-to-Site IPsec VPN tunnel between the Branch and the Cloud. Additionally, Static 1:1 NAT is configured at the Cloud edge to allow direct public access to our internal servers, while internal corporate traffic safely uses the encrypted VPN tunnel.

2 Network Topology Diagram

The network diagram below shows the 4-router setup connecting our local branch VLAN switching network to the remote cloud server farm.

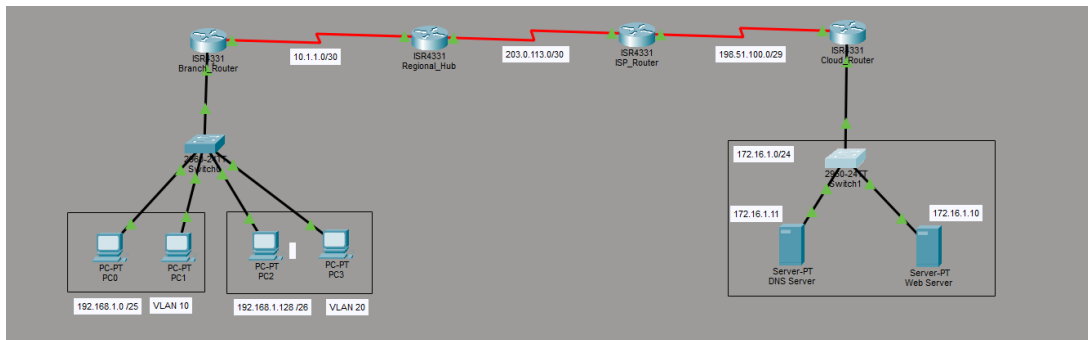


Figure 1: Enterprise Network Topology Layout

3 IP Addressing Table

We used Variable Length Subnet Masking (VLSM) to design the network subnets and optimize the allocation of IP addresses across all LAN and WAN connections.

Table 1: Network IP Addressing Space

Device Name	Interface	IP Address	Subnet Mask	Description
Branch_Router	Gi0/0/0.10	192.168.1.1	255.255.255.128	VLAN 10 Gateway (Floor 1)
	Gi0/0/0.20	192.168.1.129	255.255.255.192	VLAN 20 Gateway (Floor 2)
	Se0/1/0	10.1.1.1	255.255.255.252	Serial Link to Hub Router
Regional_Hub	Se0/1/0	10.1.1.2	255.255.255.252	Connection to Branch
	Se0/1/1	203.0.113.1	255.255.255.252	Connection to ISP
ISP_Router	Se0/1/0	203.0.113.2	255.255.255.252	Connection to Hub
	Se0/1/1	198.51.100.1	255.255.255.248	Connection to Cloud
Cloud_Router	Se0/1/0	198.51.100.2	255.255.255.248	WAN Interface (Public)
	Gi0/0/0	172.16.1.1	255.255.255.0	Cloud Inside Gateway
DNS Server	NIC	172.16.1.11	255.255.255.0	Enterprise DNS Host
Web Server	NIC	172.16.1.10	255.255.255.0	Enterprise Web Host
PC0 / PC1	NIC	DHCP Assigned	255.255.255.128	Floor 1 Devices
PC2 / PC3	NIC	DHCP Assigned	255.255.255.192	Floor 2 Devices

4 Implementation Details

4.1 VLAN and Switch Configuration

To organize the local network and isolate broadcast traffic, we configured VLANs on **Switch0**:

- **VLAN 10 (Floor 1):** Assigned to ports Fa0/1 and Fa0/2 for the first floor users.
- **VLAN 20 (Floor 2):** Assigned to ports Fa0/3 and Fa0/4 for the second floor users.
- **802.1Q Trunking:** Port Fa0/5 is set up as a trunk to pass traffic from both VLANs to the sub-interfaces on the **Branch_Router** (Router-on-a-Stick setup).

On the remote side, **Switch1** connects the cloud servers together in a flat network environment on **VLAN 1**.

4.2 DHCP Setup

The **Branch_Router** acts as the local DHCP server. We configured two separate pools: **Floor1.Pool** and **Floor2.Pool** to give dynamic IP configuration parameters to the PCs. Both pools are configured to deliver 172.16.1.11 as the primary DNS server so branch computers can resolve server names across the network.

4.3 Static NAT Configuration

We deployed **Static 1:1 NAT** on the **Cloud_Router** to map the private cloud server addresses to static public IPs so users can access them externally:

- Web Server Private IP 172.16.1.10 maps to Public IP 198.51.100.3
- DNS Server Private IP 172.16.1.11 maps to Public IP 198.51.100.4

Internal traffic going from the branch to the cloud avoids translation and enters the secure VPN tunnel directly.

4.4 Site-to-Site IPsec VPN

A secure IPsec VPN tunnel protects confidential data passing across the public ISP network between the **Branch_Router** (10.1.1.1) and the **Cloud_Router** (198.51.100.2):

- **Phase 1 (ISAKMP):** Uses Policy 10 with **AES encryption**, **SHA authentication**, and **Diffie-Hellman Group 2** managed by a static Pre-Shared Key (VPN-KEY).
- **Phase 2 (IPsec):** Uses a transform-set named TSET with **esp-aes esp-sha-hmac** parameters.
- **Traffic Selection:** Access lists (ACL 100 and ACL 101) match private traffic between the subnets (192.168.1.0/24 and 172.16.1.0/24), forcing this traffic into the encrypted tunnel.

5 Device Configuration Scripts

5.1 Branch_Router Configuration

```
ip dhcp pool Floor1_Pool
  network 192.168.1.0 255.255.255.128
  default-router 192.168.1.1
  dns-server 172.16.1.11
ip dhcp pool Floor2_Pool
  network 192.168.1.128 255.255.255.192
  default-router 192.168.1.129
  dns-server 172.16.1.11
!
crypto isakmp policy 10
  encr aes
  authentication pre-share
  group 2
crypto isakmp key VPN-KEY address 198.51.100.2
!
crypto ipsec transform-set TSET esp-aes esp-sha-hmac
!
crypto map CMAP 10 ipsec-isakmp
  set peer 198.51.100.2
  set transform-set TSET
  match address 100
!
interface GigabitEthernet0/0/0.10
  encapsulation dot1Q 10
  ip address 192.168.1.1 255.255.255.128
!
interface GigabitEthernet0/0/0.20
  encapsulation dot1Q 20
  ip address 192.168.1.129 255.255.255.192
!
interface Serial0/1/0
  ip address 10.1.1.1 255.255.255.252
  crypto map CMAP
!
ip route 0.0.0.0 0.0.0.0 10.1.1.2
access-list 100 permit ip 192.168.1.0 0.0.0.127 172.16.1.0
0.0.0.255
access-list 100 permit ip 192.168.1.128 0.0.0.63 172.16.1.0
0.0.0.255
```

5.2 Switch0 Configuration

```
interface FastEthernet0/1
  switchport access vlan 10
!
interface FastEthernet0/2
```

```

switchport access vlan 10
!
interface FastEthernet0/3
switchport access vlan 20
!
interface FastEthernet0/4
switchport access vlan 20
!
interface FastEthernet0/5
switchport mode trunk

```

5.3 Regional_Hub Configuration

```

interface Serial0/1/0
ip address 10.1.1.2 255.255.255.252
!
interface Serial0/1/1
ip address 203.0.113.1 255.255.255.252
!
ip route 192.168.1.0 255.255.255.128 10.1.1.1
ip route 192.168.1.128 255.255.255.192 10.1.1.1
ip route 172.16.1.0 255.255.255.0 203.0.113.2
ip route 198.51.100.0 255.255.255.248 203.0.113.2

```

5.4 ISP_Router Configuration

```

interface Serial0/1/0
ip address 203.0.113.2 255.255.255.252
!
interface Serial0/1/1
ip address 198.51.100.1 255.255.255.248
!
ip route 10.1.1.0 255.255.255.252 203.0.113.1
ip route 192.168.1.0 255.255.255.128 203.0.113.1
ip route 192.168.1.128 255.255.255.192 203.0.113.1
ip route 172.16.1.0 255.255.255.0 198.51.100.2

```

5.5 Cloud_Router Configuration

```

crypto isakmp policy 10
encr aes
authentication pre-share
group 2
crypto isakmp key VPN-KEY address 10.1.1.1
!
crypto ipsec transform-set TSET esp-aes esp-sha-hmac
!
crypto map CMAP 10 ipsec-isakmp

```

```

set peer 10.1.1.1
set transform-set TSET
match address 101
!
interface GigabitEthernet0/0/0
ip address 172.16.1.1 255.255.255.0
ip nat inside
!
interface Serial0/1/0
ip address 198.51.100.2 255.255.255.248
ip nat outside
crypto map CMAP
!
ip nat inside source static 172.16.1.10 198.51.100.3
ip nat inside source static 172.16.1.11 198.51.100.4
!
ip route 10.1.1.0 255.255.255.252 198.51.100.1
ip route 203.0.113.0 255.255.255.252 198.51.100.1
ip route 192.168.1.0 255.255.255.128 198.51.100.1
ip route 192.168.1.128 255.255.255.192 198.51.100.1
!
access-list 101 permit ip 172.16.1.0 0.0.0.255 192.168.1.0
0.0.0.127
access-list 101 permit ip 172.16.1.0 0.0.0.255 192.168.1.128
0.0.0.63

```

5.6 Switch1 Configuration

```

interface FastEthernet0/1
switchport mode access

```


6 Connectivity Testing and Verification

6.1 VPN Status Verification

To confirm that the IPsec tunnel is active, we generated traffic and checked the router security status using the command line interface.

```
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
198.51.100.2 10.1.1.1          QM_IDLE          1051    0 ACTIVE

IPv6 Crypto ISAKMP SA
```

Figure 2: Crypto ISAKMP SA Output Showing Active Tunnel (QM_IDLE)

6.2 End-to-End Routing Test

This test confirms connectivity between branch users and remote cloud services using ICMP packets.

```
Pinging 172.16.1.10 with 32 bytes of data:

Reply from 198.51.100.3: bytes=32 time=10ms TTL=124
Reply from 198.51.100.3: bytes=32 time=19ms TTL=124
Reply from 198.51.100.3: bytes=32 time=3ms TTL=124
Reply from 198.51.100.3: bytes=32 time=3ms TTL=124

Ping statistics for 172.16.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 19ms, Average = 8ms
```

Figure 3: Successful Ping Output From Branch Host to Server Public IP

6.3 Web Server Access Verification

This final test verifies the application tier by fetching the default company webpage via the browser client.



Figure 4: Company Webpage Loaded Successfully inside PC Browser